



Human-Aware Multi-Robot Motion Planning

MSc Thesis in Robotics

Mohammed Saleeq Kolloth

ID:2655154

Supervised by Dr Masoumeh Iran Mansouri

School of Computer Science

College of Engineering and Physical Sciences

University of Birmingham

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Abstract

Adoption of Multi-robot systems in human work environments is becoming increasingly prevalent. While this has helped to boost productivity, these systems can be made a lot safer and more efficient by better collaborative motion planning. Most existing fleet automation strategies do not consider human behavior and motion patterns into robot motion planning, but rather regard them as static or dynamic obstacles.

This project proposes a human-aware motion planning methodology for collaborative workspaces such as warehouses and factory settings. We introduce Human-aware Model Predictive Conflict-Based Search(HAMCBS), a motion planning strategy to resolve human-robot conflicts alongside other obstacles. Using a physics-based model for human motion prediction for robot planning, this CB-MPC(Conflict-Based Model Predictive Control) algorithm generates optimal trajectories.

Simulations in representative models in StageROS, as well as a custom simulator, are used to extensively assess the scalability and reliability under various scenarios. This methodology aims to serve as a starting point to advance human-centric multi-robot coordination in dynamic environments.

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Abbreviations

CBS	Conflict Based Search
CB-MPC	Conflict Based Model Predictive Control
FOV	Field Of View
HAMCBS	Human Aware Model Predictive Conflict Based search
HSA	Human Safety Area
IPOT	Interior Point Optimisation
MPC	Model Predictive Control
NLP	Non-Linear Programming
SQP	Sequential Quadratic Programming
MAPF	Multi Agent Path Finding
MARL	Multi Agent Reinforcement Learning
PSO	Particle Swarm Optimization

Contents

Abstract	ii
Acknowledgements	iii
Abbreviations	iv
List of Figures	viii
List of Tables	viii
List of Algorithms	ix
1 Introduction	1
1.1 Background	1
1.2 Motivation	2
1.3 Research Problem and Objectives	3
1.4 Contributions	3
1.5 Project structure	4
2 Literature Review	5
2.1 Human Robot collaboration	5
2.2 Multi-Robot motion planning	5
2.3 Human dynamics	6
2.4 The research Gap	6
3 Methodology: Mathematical Formulation	8
3.1 Human dynamics	8
3.2 Multi-Robot Model Predictive Control	11
3.3 Human Aware MPC formulation	12
4 Methodology: Computer simulation	14
4.1 Environmental Perception	15
4.2 Human movement prediction	15

4.3	Human-Aware planning	16
4.3.1	Global planner	16
4.3.2	Human-Aware Model Predictive Conflict-Based Search . .	18
5	Experimental Setup	21
5.1	Simulation Environment	21
5.2	Test Scenarios	22
5.3	Evaluation Metrics	22
6	Results and Discussion	25
6.1	Results	25
6.2	Comparison with Other Methods	27
6.3	Discussion	27
6.4	Limitations	29
7	Conclusion	30
7.1	Key Contributions	30
7.2	Future Work	30
	References	31
A	Instructions to run the implementaion	34

List of Figures

3.1	Figure representing various social forces influencing humans . . .	9
3.2	Planned path in horizon with new(left) vs original(right) goal tracking	12
4.1	Forming obstacles from laser scan data, starting with raw SLAM data (left) converted to occupancy map (middle) and obstacles (right)	15
4.2	Figure representing human motion prediction by the social force model with time progression from left to right	16
4.3	Discrete global planner results	18
4.5	Human and obstacle constraints in MPC horizon	19
4.4	Two robots, red and blue completing a task in the presence of a human(green). Initial MPC solution results in conflict between with blue and red robots as well as human (left). Re-planned MPC with constraints(red dots) results in a new conflict with Human(middle). Final re-planning results in conflict free solution.	19
5.1	Stage 3D simulation	21
5.2	Simulations in the Python-based custom simulator at varying scales and complexity	24
6.1	Per robot solution time vs environmental occupancy	25
6.2	Fleet makespan vs number of humans	26
6.3	Fleet makespan vs number of robots	26
6.4	Comparison of average and maximum solution time per robot as the number of humans increases.	26
6.5	Comparison of average and maximum solution time per robot as the number of robots increases.	27
6.6	Comparison of average and maximum solving time per robot in an open environment without humans.(lower is better)	27
6.7	Solving environments with high occupancy percentage	28

List of Tables

5.1	Social force model simulation parameters	22
5.2	MPC simulation parameters	23
5.3	Evaluation parameters	23
A.1	Performance metrics for different numbers of robots.	35
A.2	Performance metrics for different numbers of humans.	36

List of Algorithms

1	Human-Aware Robot Control: Main loop	14
2	Conflict-Based Search (CBS) Global Planner	17
3	Human aware Conflict Based model predictive control	20

CHAPTER 1

Introduction

1.1 Background

Automation has been on the rise in recent years, especially in manual work environments. In factories and warehouses, multi-robot fleets are commonly deployed along with a human workforce to increase throughput and reduce costs. This trend is expected to increase in the future, with some figures expecting a growth as high as 16.2%[1]. However, with this increase in human-robot collaborative workspaces, there has been a pattern of increasing workplace accidents and a productivity loss due to human reaction, psychological factors, and the inability of robots to account for the same [2]. On the contrary, studies have established a clear increase in productivity with collaborative robots, and suggested that human-aware planning can objectively provide a definite advantage [3]. This work aims to address the human-robot collaboration problem at the robot level by introducing human awareness to the robot fleet.

The fleet automation problem is a classic problem in robotics has been extensively studied. One such problem can be divided into five different sub-problems, namely,[4]

1. **Task allocation:** Given area constraints and tasks to be performed with mission requirements, assign objectives for each robot.
2. **Motion planning:** Estimate vehicle path, given goals, vehicle kinematics, and feasibility of control and position.
3. **Coordination:** Given control constraints and motion plan, compute time synchronisations to avoid collision.
4. **Temporal reasoning:** Calculate velocities given motion plan, synchronisation, and deadlines.
5. **Control:** Execute vehicle motions given reference paths, velocities, and feasibility constraints.

We can use the same principles to break down and explain the situation of getting a fleet of robots to work with human awareness. This project aims to develop a human-aware model for multi-robot motion planning, coordination, and finally, temporal reasoning and control, to enhance the safety and efficiency of such systems.

1.2 Motivation

Human-aware motion planning is crucial in industrial multi-robot systems due to several critical issues that arise when human presence is not adequately considered. These issues can be summarized as follows:

1. **Safety Concerns:** Without proper consideration of human presence and behavior, robots may pose significant safety risks. Accidents can occur due to unexpected human actions or the robots' inability to predict and adapt to these actions in real-time.
2. **Reduced Efficiency:** When robots cannot effectively anticipate and respond to human movements, productivity is often compromised. Robots may have to slow down, stop, or unnecessarily reroute, leading to delays and inefficiencies in the workflow.
3. **Psychological Impact on Human Workers:** The presence of robots in close proximity can cause stress and discomfort among human workers, particularly if the robots' movements are perceived as unpredictable or unsafe. This can lead to reduced job satisfaction, lower productivity, and even resistance to adopting robotic systems.

While robot fleets in industrial settings are not a new concept, existing attempts to address the integration of human awareness into these systems have been somewhat limited. On the task allocation front, models that leverage human awareness, such as those using game theory, have been studied [5]. However, there remains a gap in incorporating human awareness into subsequent steps of the fleet automation process.

Most contemporary systems adopt well-established Multi-Agent Path Finding (MAPF) algorithms like M* (M-star) [6], Conflict-Based Search (CBS) [7], and Particle Swarm Optimization (PSO) [8], treating humans as dynamic or static obstacles during the planning phase. Although these approaches address collision concerns, they fail to account for human motion, leading to frequent re-planning and sudden maneuvers that negatively affect interactions with human co-workers.

In most recent research, advancements have been made in navigation for warehouse environments for single-robot motion planning, utilizing techniques such as Human Safety Area (HSA) fusion[9]. Additionally, modern approaches like attention-based deep Multi-Agent Reinforcement Learning (MARL) have been explored, focusing on isolated cases of robot-crowd interaction in obstacle-free spaces[10]. However, these methods prove to be incompatible with highly randomized environments like warehouses and lack scalability—an essential factor in such systems. Moreover, learning-based models often fail to generate consistent results for identical problem configurations and cannot guarantee the existence of a solution, making them less adaptable for practical use.

A motion planning algorithm capable of leveraging human dynamics and addressing the concerns outlined in the previous section remains a significant gap in current research. Although substantial research exists on human movement in various environments [11], this project aims to bridge this gap and introduce a human-centred collaborative motion planner increasing safety and efficiency.

1.3 Research Problem and Objectives

The research problem addressed here is the development of a motion planning algorithm that integrates human awareness into multi-robot systems to enhance safety and efficiency in dynamic industrial environments.

Key Objectives in this study are,

1. **Generate Optimal Paths:** Develop algorithms that produce optimal paths considering both robot and human dynamics.
2. **Handle Coordination:** Implement coordination strategies to avoid collisions between robots and with humans.
3. **Ensure Temporal Feasibility:** Ensure the proposed paths and plans adhere to temporal constraints.
4. **Generate Control Parameters:** Develop control parameters that guide robot movements effectively.

The scope of this study is limited to randomized environments with dynamic obstacles limited to humans and other robots the environmental elements static during planning. The algorithm will be tested in various randomized scenarios to validate its effectiveness in handling different configurations and ensuring no violations of human safety areas.

1.4 Contributions

The most important contribution of this work is the development of the Human-aware model predictive Conflict-based search (HAMCBS) algorithm. HAMCBS was developed by introducing,

- Dynamic constraint addition
- Human dynamic constraints based on the social force model [12] and HSA fusion [9]
- A modified objective function for goal tracking to the original CB-MPC, making handling dynamic obstacles and human motion prediction possible.

This work also presents a thorough study on the scalability and reliability of the developed algorithm by testing it with varying amounts of humans, robots, and environmental complexity. This work would serve as a starting point to a variety of human-aware planning methods, enhancing the safety and efficiency of a human-robot collaborative workspace.

1.5 Project structure

This thesis is organized into six chapters to systematically address the research objectives outlined above.

Chapter 1: Introduction

Provides an overview of the research topic, including the background, motivation, research problem, and objectives. It also outlines the contributions of this study and presents the structure of the thesis.

Chapter 2: Literature Review

Examines existing research related to multi-robot motion planning and human-robot interaction. It identifies key theories, models, and gaps in the current literature that this thesis aims to address.

Chapter 3: Methodology:Mathematical Formulation

Describes how the proposed human dynamics and motion planning model is theoretically formulated

Chapter 4:Methodology:Computer simulation

Outlines the implementation process of the algorithm from mathematical foundations.

Chapter 5:Experimental setup

Describes the simulation environment, software tools, and evaluation metrics.

Chapter 5: Results and Analysis

Presents the experimental results obtained from testing the algorithm in various simulated environments. The chapter analyzes the performance of the algorithm in terms of efficiency and scalability.

Chapter 6: Conclusion and Future Work

Summarizes the key findings of the research, discusses their implications, and suggests directions for future research to further enhance human-aware motion planning in multi-robot systems.

CHAPTER 2

Literature Review

Understanding the current literature related to Human-Aware multi-robot motion planning has greatly helped in addressing the proposed research question of developing a human-aware multi-robot motion planner. Research in this area can be thematically classified into Human-robot collaboration, Multi-robot planning, and Human dynamics.

2.1 Human Robot collaboration

One of the prime questions to address with the proposed research question is whether human-aware planning has any advantages at all, and if so, how significant they are. Lasota et al. [3] conducted a study, examining the effectiveness of human-aware motion planning in close proximity human-robot interaction. The study concluded that significant benefits can be achieved from adapting human awareness in the motion planning level, improving teamwork and worker satisfaction. Kenk et al's attempted the problem from a single robot perspective, where they implemented a human-aware navigation system for warehouse environments[9]. The idea of identifying humans and performing an online re-planning to keep out of the HSA was notable in this work.

A few learning based solutions were also proposed for similar problems. Dong et al. proposed a socially aware multi-robot model using attention-based deep MARL. Despite the many fundamental differences, this work also estimates a social feature and uses it as an input to a MARL path planner, much like the methodology proposed in this thesis, estimating a social motivation Langvian equation to be integrated into a MAPF solver.

2.2 Multi-Robot motion planning

Multi-agent path finding (MAPF) problem has a variety of well-established solutions, with some having the ability to deal with dynamic obstacles. Particle

Swarm Optimization[8], Gravitational search[13], Multi Agent Reinforcement Learning(MARL)[10] and Conflict-Based Search(CBS)[7] are considered here. PSO and Gravitational search work almost in a similar way with a Bayesian filter retaining only the most fit solutions. These often get stuck in local minima and are not deterministic. In an environment like a factory or warehouse, a deterministic planner would be more appropriate so that the task planner can make definite decisions. This issue is present with MARL as well, on top of requiring learning from a highly randomized environment. CB-MPC, a subset of CBS with model predictive control[14], is adapted here. CBS has been extensively modified before, with some works suggesting dynamic constraint addition[15]. Due to its versatility and deterministic nature, a modified CB-MPC built on the original work of Tajbaksh et al.[14], modified with the suggestion of adopting it to handle dynamic obstacles as proposed in the ICBS(Improved CBS)[15], was considered a good starting point.

2.3 Human dynamics

Human motion prediction is a well-explored area. Most of the methods in this area can be classified into three different categories [11]. First among these are physics-based models. Physics-based models include the various social force models proposed by Helbing et al.[12], Moussaïd et al.[16]. These models can be readily applied in any environment. However, they carry a limitation of decreasing accuracy over time, making them usable only in a short prediction horizon.

The other models are pattern-based and planning-based. Pattern-based methods learn from data and can make complex predictions. Requiring data in a specific environment limits these models. These include models like the social LSTM [17].

Planning-based approaches are similar to robot planning, with physics-based models in the core [11]. They have good performance, given a map and goals in a given environment; however, they cannot handle obstacles or changing maps.

Considering environmental flexibility, data limitations, and acceptable approximations from previous studies examining robots in crowded settings [10], we adopt a physics-based approach using social forces for this work. It is also to be noted that since we only need the human model to be accurate in the short MPC horizon, a physics-based model is a justified choice.

2.4 The research Gap

To accomplish human-aware multi-robot motion planning, our intended planning and control strategy should be able to predict human dynamics, plan optimal paths for multiple robots, and resolve collisions and conflicts. Existing human-robot collaborative planners fail to meet all of the objectives simultaneously. Most of the methodologies regard the human presence as a static or dynamic obstacle, with PSO[8] assigning random motion. This greatly reduces the optimality of planning, resulting in unnecessary re-routing and sudden or unnatural movements disrupting human activity. A motion planning solution with social awareness integrated at the local planning level remains a gap that

this work attempts to address.

CHAPTER 3

Methodology: Mathematical Formulation

In this section, we will examine the mathematical formulation of the human movement prediction based on the social force model, preliminary constraints defining collisions, and the specific Model Predictive Control objective to estimate an appropriate control strategy in a given horizon.

3.1 Human dynamics

Prediction of human dynamics is of prime importance to achieve human-aware motion planning. The physics-based social force model, originally proposed by Helbing et al.[12], is adapted in this work. The social force model works on the principle of calculating human motion much like a physical interaction, where social forces represent the motivation of an individual to take a particular action at any given moment, rather than a force that moves the person directly. On a higher level social force model can be decomposed into three component force terms.

1. Acceleration towards the desired direction of motion.
2. Repulsive forces from other humans and borders.
3. Attractive forces towards friendly or distracting targets.

For the human under consideration α , we can define the position vector of the person at any given time t as $\vec{r}_\alpha(t)$. Assuming the person is trying to get to objective o at position vector $\vec{r}_\alpha^o$, we make an assumption to simplify the situation that the person would want to take the shortest route, represented by a set of waypoints with the final way point at $\vec{r}_\alpha^o$. We will represent the very next way point by $\vec{r}_\alpha^k$. Now, the direction the human will take in the next step $\vec{e}_\alpha(t)$ can be estimated as [12],

$$\vec{e}_\alpha(t) = \frac{\vec{r}_\alpha^k - \vec{r}_\alpha(t)}{\|\vec{r}_\alpha^k - \vec{r}_\alpha(t)\|} \quad (3.1)$$

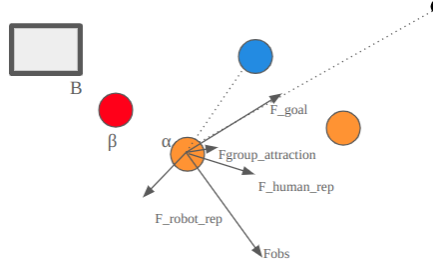


Figure 3.1: Figure representing various social forces influencing humans

A second assumption is made at this point, that the person has a desired speed for a given objective represented by v_α^o , to which they will default. The velocity of the person can now be expressed as [12],

$$\vec{v}_\alpha^o(t) = v_\alpha^o \vec{e}_\alpha(t) \quad (3.2)$$

However, due to external interactions, the human would need to slow down or speed up throughout the course of motion. The tendency of the person to default to the desired velocity would hence create a motivation force, $\vec{F}_\alpha^o$ can be calculated based on the current velocity of the person $\vec{v}_\alpha$ and the desired velocity $\vec{v}_\alpha^o(t)$ as,[12]

$$\vec{F}_\alpha^o(\vec{v}_\alpha, \vec{v}_\alpha^o(t)) = \frac{1}{\tau} (v_\alpha^o \vec{e}_\alpha - \vec{v}_\alpha) \quad (3.3)$$

The force in equation 3.3 represents the first term of our social force model. The second term spans all the repulsive forces, including other humans and obstacle borders. Interpersonal interactions are defined based on a model of sphere of influence in the social force model. The repulsive force from another human increases closer to them. This is expressed by creating a monotonically decreasing potential field, as a function of an apparent relative distance b , calculated to represent closeness while considering the stepping distance of the human and the direction of motion of the other human. Between the person under consideration α and a human β , this apparent distance is defined as per equation 3.4 below.

$$b = \frac{1}{2} \sqrt{(\|\vec{r}_{\alpha\beta}\| + \|\vec{r}_{\alpha\beta} - v_\beta \Delta t \vec{e}_\beta\|)^2 - (v_\beta \Delta t)^2} \quad (3.4)$$

The resulting potential field $V_{\alpha\beta}$ will have potential lines in the shape of ellipses, with a minor axis of length $2b$. The major axis of the ellipse points towards the direction of motion, with its length depending on the movement speed.

Force exerted by this potential field $\vec{f}_{\alpha\beta}$ can be estimated by simply taking a gradient with respect to $\vec{r}_{\alpha\beta}$.

$$\vec{f}_{\alpha\beta}(\vec{r}_{\alpha\beta}) = -\nabla_{\vec{r}_{\alpha\beta}} V_{\alpha\beta}[b(\vec{r}_{\alpha\beta})] \quad (3.5)$$

Repulsive force from obstacle borders is estimated in a similar way by defining a monotonically decreasing potential field $U_{\alpha B}$. Here, the potential can be estimated simply by considering the relative distance to the closest point

in the border B , represented by $\vec{r}_{\alpha B}$. The border induces a social force $\vec{F}_{\alpha B}$, formulated in equation 3.6 below.

$$\vec{F}_{\alpha B} = -\nabla_{\vec{r}_{\alpha B}} U_{\alpha B}(\|\vec{r}_{\alpha B}\|) \quad (3.6)$$

Attractive motivation forces make up the last part of the social force model. At a fixed point in time, the formulation of these forces are similar to what we discussed above, except that an attractive potential is utilized. For a point of distraction interest i , the potential field for human α is defined as $W_{\alpha i}$, a monotonically increasing positive potential. However, it is to be noted that humans lose interest over time. To capture this effect, $W_{\alpha i}$ is a multivariable function with one of the independent variables being time. The maximum attractive force $\vec{f}_{\alpha i}$ can be defined as,

$$\vec{f}_{\alpha i} = -\nabla_{\vec{r}_{\alpha i}} W_{\alpha i}(\|\vec{r}_{\alpha i}\|, t) \quad (3.7)$$

where $\vec{r}_{\alpha i}$ is the relative position vector of distraction i from α .

Except for the goal motivation $\vec{F}_{\alpha}^0$, all of the other social forces are able to influence the humans movement decision making process when the other human subject β , the border B and or the distraction i are perceived by the them. Assuming human perception is strongest in the field of view (FOV), a weighting function needs to be applied to the respective force terms to model the perception effect.

Defining a total FOV of 2ϕ , we can define an influence coefficient c , where $c \in (0, 1)$. Now we can define the weighting function w as a piece-wise function of the force vector and the desired direction vector.

$$w(\vec{e}, \vec{f}) = \begin{cases} 1 & \text{if } \vec{e} \cdot \vec{f} \geq \|\vec{f}\| \cos \phi \\ c & \text{otherwise} \end{cases} \quad (3.8)$$

Now, using equations 3.5, 3.6, and 3.7, we can define the final form of the motivation forces.

$$\vec{F}_{\alpha\beta}(\vec{e}_{\alpha}, \vec{r}_{\alpha\beta}) = w(\vec{e}_{\alpha}, -\vec{f}_{\alpha\beta}) \vec{f}_{\alpha\beta}(\vec{r}_{\alpha\beta}) \quad (3.9)$$

$$\vec{F}_{\alpha B}(\vec{e}_{\alpha}, \vec{r}_{\alpha B}) = w(\vec{e}_{\alpha}, -\vec{f}_{\alpha B}) \vec{f}_{\alpha B}(\vec{r}_{\alpha B}) \quad (3.10)$$

$$\vec{F}_{\alpha i}(\vec{e}_{\alpha}, \vec{r}_{\alpha i}, t) = w(\vec{e}_{\alpha}, \vec{f}_{\alpha i}) \vec{f}_{\alpha i}(\vec{r}_{\alpha i}, t) \quad (3.11)$$

As shown in Figure 3.1, a vector sum of all motivation forces from all obstacle borders, humans, and distractions can provide us with the net movement motivation for human α

$$\vec{F}_{\alpha}(t) = \vec{F}_{\alpha}^0(\vec{v}_{\alpha}, \vec{v}_{\alpha}^o(t)) + \sum_{\beta} \vec{F}_{\alpha\beta}(\vec{e}_{\alpha}, \vec{r}_{\alpha\beta}) + \sum_B \vec{F}_{\alpha B}(\vec{e}_{\alpha}, \vec{r}_{\alpha B}) + \sum_i \vec{F}_{\alpha i}(\vec{e}_{\alpha}, \vec{r}_{\alpha i}, t) \quad (3.12)$$

It is imperative to note that the social force is just defining a human's motivation to move, not the movement itself. Human motion is still stochastic, hence the final desired velocity of human $w_{\alpha}(t)$ can be modelled using a Langevin equation, whose deterministic part comes from the motivation force and non-deterministic part defined by a stochastic distribution η . Mathematically, we can represent this as,

$$\frac{d\vec{w}_{\alpha}}{dt} = \vec{F}_{\alpha}(t) + \eta(t) \quad (3.13)$$

This value could sometimes be physically impossible to achieve for the human due to the arbitrary nature of being derived from the motivation itself. In order to predict the actual human velocity, we can simply limit the maximum magnitude to the upper bound of humans physically capable speed v_α^{max} . This could simply be represented by a piecewise function as shown below.

$$\vec{v}_\alpha(t) = \begin{cases} \vec{w}_\alpha(t) & \text{if } \|\vec{w}_\alpha\| \leq v_\alpha^{max} \\ \vec{w}_\alpha(t) \frac{v_\alpha^{max}}{\|\vec{w}_\alpha(t)\|} & \text{otherwise} \end{cases} \quad (3.14)$$

The resulting Langvian equation has been evaluated to be able to reasonably predict human motion.[12][16] Human positions estimated from this model is used in the further stages of this project to predict possible human position distribution.

3.2 Multi-Robot Model Predictive Control

Model predictive control is an advanced control strategy used in systems with multiple control variables and constraints[18]. MPC is a receding horizon control strategy with continuous updating and adaptability in each time step. This would make it ideal in a highly dynamic system like ours with stochastic constraints concerning humans.

The bare bones MPC to control a robot with a kinematic model f to move in a given path defined by reference states r to reach a desired state x_d withing a tolerance of ϵ_g can be described with the following constraints 3.15 through 3.18[18].

$$x^{k+1} = f(x^k, u^k) \quad (3.15)$$

$$\lim_{k \rightarrow \infty} \|x^d - x^k\| \leq \epsilon_g \quad (3.16)$$

$$x^k \in X_0 \quad \text{and} \quad x^{k+N} \in X_f \quad (3.17)$$

$$x^l \in X_{\text{feasible}} \quad \text{and} \quad u^l \in U_{\text{feasible}} \quad (3.18)$$

Where N is the length of the planning horizon, x, u state and control, respectively, and l represents the time instant in the planning horizon. The control signal u and resulting state x can be obtained by minimizing the objective J in the control horizon as given below,

$$\text{argmin}_{u,x} J = \sum_{l=k}^{k+N-1} (x^l - r^l)^T Q (x^l - r^l) + u^{l,T} R u^l + x^{k+N,T} P x^{k+N} \quad (3.19)$$

Here Q, R, P are weighing matrices to determine the relative importance of reference tracking, control effort and goal tracking. Tajbaksh et al. proposed a conflict-based model predictive control strategy to control multi-robot systems in static environments.[14] This model added to this basic MPC formulation to handle static obstacles and inter-robot collision avoidance. This was made possible by introducing the following additional constraints to the system.

$$\|p_j^k - p_i^k\| \geq D + \delta_r + \epsilon_r \quad \forall i \in N_a \quad (3.20)$$

$$\|p_o^p - p_i^k\| \geq D/2 + \delta_o + \epsilon_o \quad \forall i, j \in N_a \quad (3.21)$$

where N_a is the number of robots, D is the robot footprint. Tolerances for collision with robots and static obstacles are defined as ϵ_r and ϵ_o , respectively. Additionally, the objective for each robot i was redefined to allow soft violation of these constraints with slacks δ_r, δ_o , and slack coefficients k_r, k_o , respectively.

$$\operatorname{argmin}_{u_i, x_i, \delta_r, \delta_o} J_i = \sum_{l=k}^{k+N-1} (x_i^l - r_i^l)^T Q (x_i^l - r_i^l) + u_i^{l,T} R u_i^l + x_i^{k+N,T} P x_i^{k+N} + k_r \delta_r + k_o \delta_o \quad (3.22)$$

It is imperative that these slacks be non-negative to avoid collisions in cases with constraint violation.

$$\delta_r, \delta_o \geq 0 \quad (3.23)$$

3.3 Human Aware MPC formulation

Human-Aware Model Predictive Conflict-based search is in its core, utilizes a model similar to the Multi-robot MPC model discussed above. The formulation is derived by introducing the following core changes.

1. Defining the goal tracking term to minimise the horizon end state to the desired end state in each step.
2. Addition of preliminary constraint for human collision.
3. Integrating violation cost to human collision in the objective function.

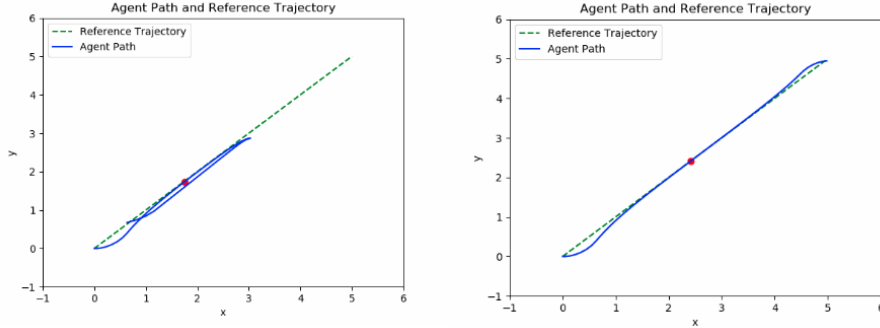


Figure 3.2: Planned path in horizon with new(left) vs original(right) goal tracking

The resulting formulation can be defined based on the following preliminary constraints for a system of N_a robots and N_h humans:

$$x_i^{k+1} = f(x_i^k, u_i^k) \quad (3.24)$$

$$\lim_{k \rightarrow \infty} \|x_i^d - x_i^k\| \leq \epsilon_g \quad (3.25)$$

$$\|p_j^k - p_i^k\| \geq D + \delta_r + \epsilon_r \quad \forall i \in N_a \quad (3.26)$$

$$\|p_o^k - p_i^k\| \geq D/2 + \delta_o + \epsilon_o \quad \forall i, j \in N_a \quad (3.27)$$

$$\|p_j^h - p_i^k\| \geq D_{HSA} + \delta_h \quad \forall j \in N_h, \forall i \in N_a \quad (3.28)$$

Here, the newly introduced symbols D_{HSA}, δ_h, k_h represent human safety area diameter, the slack for violating HSA, and the penalising multiplier for the violation, respectively.

The objective for each robot can now be defined as,

$$\begin{aligned} \underset{u_i, x_i, \delta_r, \delta_o, \delta_h}{\operatorname{argmin}} J_i = & \sum_{l=k}^{k+N-1} \left[(x_i^l - r_i^l)^T Q (x_i^l - r_i^l) + u_i^{l,T} R u_i^l \right] \\ & + (x_i^{k+N,T} - x_i^d)^T P (x_i^{k+N,T} - x_i^d) \\ & + k_r \delta_r + k_o \delta_o + k_h \delta_h \end{aligned} \quad (3.29)$$

This minimisation is subject to,

$$x_i^k \in X_0 \quad \text{and} \quad x_i^{k+N} \in X_f \quad (3.30)$$

$$x_i^l \in X_{\text{feasible}} \quad \text{and} \quad u_i^l \in U_{\text{feasible}} \quad (3.31)$$

$$\delta_r, \delta_o, \delta_h \geq 0 \quad (3.32)$$

The new objective would be able to plan continuous robot paths between horizons as shown in the simulation results in Figure 3.2. This is necessary as this would make conflict resolution solvable in the algorithm.

CHAPTER 4

Methodology: Computer simulation

This chapter describes the details of how human-aware motion planning is achieved based on the mathematical formulations developed. As shown in Algorithm 1 the general order of motion plan estimation is to,

1. Get the initial environment state with humans, robots, obstacles, and distractions.
2. Predict human movement for a short time horizon.
3. Estimate controls for all robots of the fleet.
4. Update the robot and human positions.
5. repeat till task is finished.

Algorithm 1 Human-Aware Robot Control: Main loop

```
1: Input: Initial human and robot states and goals  $S_r, S_h, G_r, G_h$ , static obstacles  $O$ 
2: Output: Executed robot movements at every time  $t$  based on control  $U_t$ 
3:  $\rho \leftarrow \text{Generate\_reference}(O, S_r, S_g)$ 
4:  $MPC \leftarrow \text{new HAMCBS}(O, \rho)$ 
5:  $X_r \leftarrow S_r$ 
6: for  $t = 0$  to  $T_{\max}$  do
7:   if  $\|S_r^k - G_r\| < \epsilon_g$  then
8:     break
9:   end if
10:   $X_h \leftarrow \text{Get\_Human\_states}(S_h, G_h, O)$ 
11:   $M \leftarrow (O, X_r, G_r, X_h, N)$ 
12:   $U_t \leftarrow MPC.\text{Conflict\_Solve}(M)$ 
13:   $X_r \leftarrow \text{Move\_and\_Update\_states}(U_t)$ 
14: end for
15: Return  $A$ 
```

Here, ρ is the matrix of reference paths, $\rho = [r_0, r_1, \dots, r_{N_a-1}]^T$. Also, X_r and X_h represent the current positions of robots and humans, respectively. We can now examine the details of all the methods adopted.

4.1 Environmental Perception

Understanding the obstacles in a robot's environment is crucial for accurately estimating both robot and human movements. In the social force model, potential fields are calculated for each boundary, with every obstacle exerting a force on the moving entities.

In this project, to closely align with real-world applications, point cloud data from Simultaneous Localization and Mapping (SLAM) is used. However, directly adopting this data would make the solution computationally infeasible due to the vast number of points and the corresponding forces that would need to be calculated. This challenge is similarly present in the Model Predictive Control (MPC) subroutine, where adding constraints from obstacles requires the identification of individual obstacles to simplify the problem.

To address this, laser scan data is converted into a binary occupancy map. The Hough Transform [19] is employed to extract boundaries as lines, simplifying the environment representation. Figure 4.1 illustrates this process, showing the progression from raw SLAM data (left) to the binary occupancy map (middle) and finally to the extracted obstacles (right).

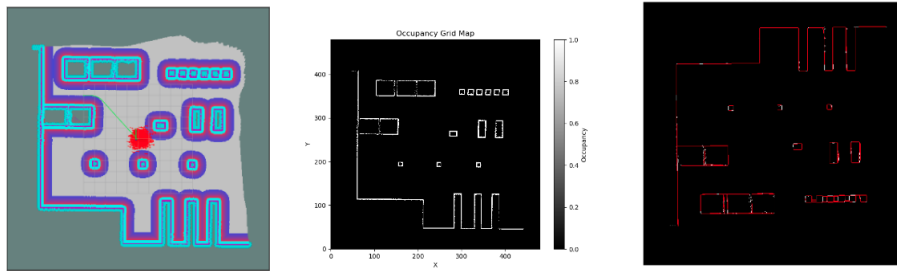


Figure 4.1: Forming obstacles from laser scan data, starting with raw SLAM data (left) converted to occupancy map (middle) and obstacles (right)

4.2 Human movement prediction

Since we have identified obstacles, to implement human movement prediction, only the calculation of the potential fields formulated in equations 3.5, 3.6 in chapter 3 remains. We can recall the motivation forces from other humans $\vec{f}_{\alpha\beta}(\vec{r}_{\alpha\beta})$ and obstacle borders $\vec{F}_{\alpha B}$ respectively as

$$\vec{f}_{\alpha\beta}(\vec{r}_{\alpha\beta}) = -\nabla_{\vec{r}_{\alpha\beta}} V_{\alpha\beta}[b(\vec{r}_{\alpha\beta})]$$

$$\vec{F}_{\alpha B} = -\nabla_{\vec{r}_{\alpha B}} U_{\alpha B}(\|\vec{r}_{\alpha B}\|)$$

The potentials $V_{\alpha\beta}$ and $U_{\alpha B}$ are to be defined. To conform with the expected behaviour, these terms had to be defined as monotonically decreasing repulsive.

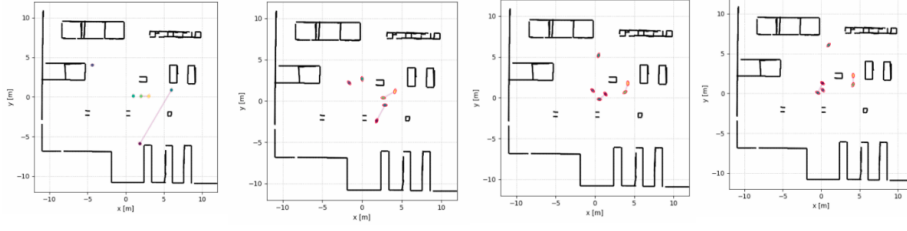


Figure 4.2: Figure representing human motion prediction by the social force model with time progression from left to right

An exponential function with negative power was selected, following the example of the original work[12].

$$V_{\alpha\beta} = V_{\alpha\beta}^0 e^{-b/\sigma} \quad (4.1)$$

$$U_{\alpha B} = U_{\alpha B}^0 e^{-\|\vec{r}_{\alpha B}\|/R} \quad (4.2)$$

Here σ is the standard deviation of the other pedestrian velocities, and b is the apparent distance we defined during formulation. The radius R represents the radius of an ellipse, effectively conformed to a circle by having equal major and minor axes due to the effect of the border. The diameter of such a circle would equal to actual speed of the human $\|\vec{v}_{\alpha}\|$.

4.3 Human-Aware planning

With the physics-based social force model implemented in the previous section, we can now produce human possible positions accurately in a short horizon. The human awareness component in this work is integrated into the MPC local planner, as we discussed in Chapter 3, sections 2 and 3.

To solve the underlying Multi-Agent Path Finding (MAPF) problem, we will need a global as well as a local planner implementation.

4.3.1 Global planner

To create motion plans for given tasks with human awareness, we are adopting a rudimentary conflict-based search (CBS) global planner. The global plan can produce a conflict-free and collision-free path as a starting point, which would make the local planning problems computationally less complex.

The global planner discretizes the environment, and uses a low-level planner such as A*(A-star) to plan out individual robot paths as illustrated in Figure 4.3(a). Each combination of robots is then conflict checked to ensure that no two robots occupy the same cell at the same time step. In the event that there is a conflict, the cell is constrained for one of the robots, and its path is replanned. This process repeats until no conflicts remain, as illustrated in Figure 4.3(b).

The detailed algorithm for the global plan is given below in Algorithm 2,

Algorithm 2 Conflict-Based Search (CBS) Global Planner

```

1: Input: Initial robot states and goals  $S_r, G_r$ 
2: Output: Conflict-free reference paths for all robots  $\rho$ 
3: conflict_found  $\leftarrow$  True
4:  $\rho \leftarrow []$ 
5: for  $i \leftarrow 0$  to  $N_a - 1$  do
6:   robot  $\leftarrow S_r[i]$ 
7:   start  $\leftarrow$  robot[0]
8:   goal  $\leftarrow G_r[i]$ 
9:   path  $\leftarrow$  Grid_env.A_star(start, goal)
10:   $\rho.append(path)$ 
11: end for
12: n_itr  $\leftarrow 0$ 
13: while conflict_found or n_itr < 1000 do
14:   conflict_found  $\leftarrow$  False
15:   n_itr  $\leftarrow$  n_itr + 1
16:   for  $i \leftarrow 0$  to  $N_a - 1$  do
17:     path1  $\leftarrow \rho[i]$ 
18:     for  $j \leftarrow 0$  to  $i - 1$  do
19:       path2  $\leftarrow \rho[j]$ 
20:       for step  $\leftarrow 0$  to  $\min(\text{len}(\text{path1}), \text{len}(\text{path2})) - 1$  do
21:         if  $\|\text{path1}[\text{step}] - \text{path2}[\text{step}]\| < \epsilon_r$  then
22:           conflict_found  $\leftarrow$  True
23:           conflict_time  $\leftarrow$  step
24:           conflict_pos  $\leftarrow$  path1[step]
25:            $x, y \leftarrow$  conflict_pos
26:           occupancy  $\leftarrow$  Grid_env.occupancy_grid
27:           Grid_env.block_range(x, y, radius = D)
28:           path  $\leftarrow$  Grid_env.A_star(start, goal)
29:            $\rho[i] \leftarrow$  path
30:           Grid_env.occupancy_grid  $\leftarrow$  occupancy
31:           break
32:         else
33:           conflict_found  $\leftarrow$  False
34:         end if
35:       end for
36:     if conflict_found then
37:       break
38:     end if
39:   end for
40: end for
41: end while

```

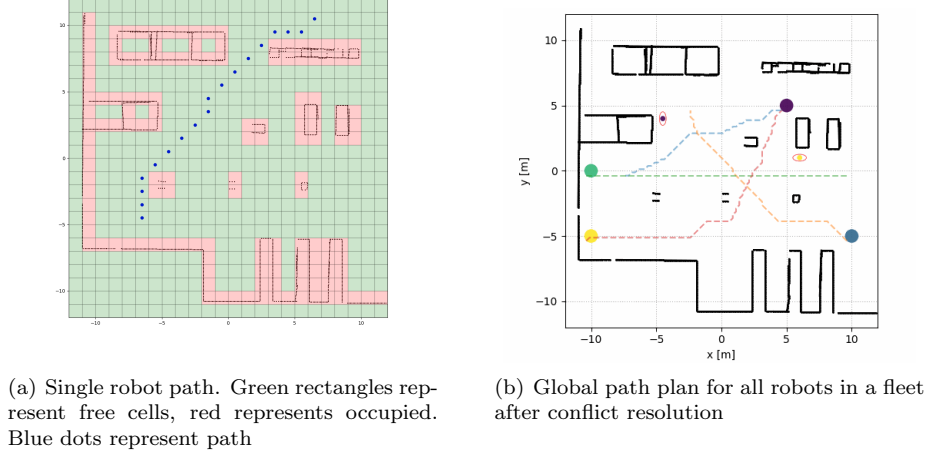


Figure 4.3: Discrete global planner results

4.3.2 Human-Aware Model Predictive Conflict-Based Search

Figure 4.4 presents an example scenario for the local planner. Given the reference states from the global planner, the Human-aware Model Predictive Control (MPC) algorithm formulated in chapter 3, determines a trajectory for each robot in the fleet while adhering to constraints, as detailed in lines 6-11 of Algorithm 3. For each solution trajectory, the cost is evaluated as the sum of the individual costs for all agents, calculated as the sum of the trajectory length and the distance to the desired state:

$$\sum_{N_a} \text{Length}(\mathbf{x}_i^N) + |x_i^{k+N-1} - x_i^d| \quad (4.3)$$

These solutions are then inserted into an array that is sorted based on the cost. In each iteration, the lowest-cost node is extracted. If this node has no conflicts, it is returned as the solution, as described in lines 15-17.

In case of a conflict, such as the one depicted in our illustrative example, the algorithm considers the first conflict, which, in this case, is an inter-robot collision. The algorithm then generates new solutions by alternately constraining both agents involved and adds these solutions to the solution tree. Since the tree is self-sorting, the lowest-cost solution will be selected next.

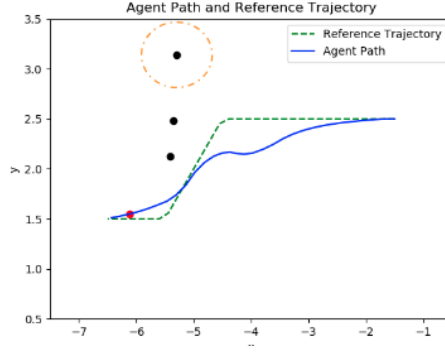


Figure 4.5: Human and obstacle constraints in MPC horizon

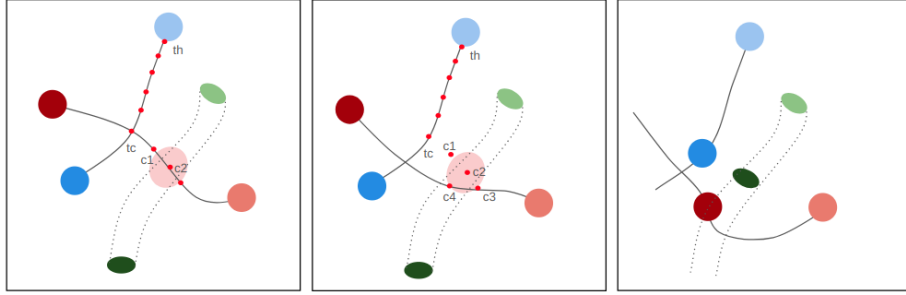


Figure 4.4: Two robots, red and blue completing a task in the presence of a human(green). Initial MPC solution results in conflict between with blue and red robots as well as human (left). Re-planned MPC with constraints(red dots) results in a new conflict with Human(middle). Final re-planning results in conflict free solution.

If a conflict persists, such as a violation of the human safety area, additional constraints are applied, and the process repeats until a conflict-free solution is found. Figure 4.5 illustrates how the MPC resolves a given problem by applying constraints within the prediction horizon. This necessitates redefining the MPC problem at each horizon, as the constraints—and therefore the problem itself—change dynamically with each iteration.

The MPC problem is modelled as a Non-Linear Programming problem (NLP). In each iteration of the receding horizon, the additional constraints are checked and based on that, new parameters are dynamically added for obstacles, humans, and robots as computed in lines 25,33,47 of Algorithm 3

Algorithm 3 Human aware Conflict Based model predictive control

```

1: Input:  $M(O, X_r, G_r, X_h, N)$ 
2: Output: Collision-Free Trajectories  $P.solution$ 
3:  $R \leftarrow \text{Node}(\epsilon_r, \epsilon_o, D)$ 
4:  $R.humans \leftarrow X_h$ 
5:  $T_s \leftarrow \text{Solution\_Tree}()$ 
6: for  $a_i \leftarrow 0$  to  $N_a - 1$  do
7:    $X_r \leftarrow \text{Get\_pose}(S_g)$ 
8:    $O_h \leftarrow \text{Get\_obstacles\_in\_range}(X_r, O)$ 
9:    $R.obstacles[a_i] \leftarrow O_h$ 
10:   $R.solution[a_i] \leftarrow \text{MPC}(M, a_i, \{\})$ 
11: end for
12:  $R.cost \leftarrow \text{SIC}(R.solution, G_r)$ 
13:  $T_s.insert\_node(R)$ 
14: while True do
15:    $P \leftarrow T_s.\text{Get\_lowest\_cost\_solution}()$ 
16:   if  $P.conflicts() == 0$  then
17:     return  $P.solution$ 
18:   end if
19:    $C \leftarrow P.conflicts()[0]$  ▷ Get the first conflict
20:    $a_i, a_j \leftarrow C.agents$ 
21:   if  $\text{Conflict\_type}(a_i, a_j)$  is Obstacle then ▷ Obstacle Collision
22:      $A \leftarrow \text{Node}(\epsilon_r, \epsilon_o, D)$ 
23:      $A.obstacles \leftarrow P.obstacles$ 
24:      $A.constraints[a_i].append(\text{Constraint\_type}, A.obstacles[a_i], C.collision\_timesteps)$ 
25:      $A.solution[a_i] \leftarrow \text{MPC}(M, a_i, A.constraints[a_i])$ 
26:      $A.cost \leftarrow \text{SIC}(A.solution, G_r)$ 
27:      $T_s.insert\_node(A)$ 
28:   else if  $\text{Conflict\_type}(a_i, a_j)$  is Human then ▷ Human Collision
29:      $A \leftarrow \text{Node}(\epsilon_r, \epsilon_o, D)$ 
30:      $A.obstacles \leftarrow P.obstacles$ 
31:      $A.humans \leftarrow P.humans$ 
32:      $A.constraints \leftarrow P.constraints$ 
33:      $A.constraints[a_i].append(\text{Constraint\_type}, \text{Get\_pose}(A.humans), C.collision\_timesteps)$ 
34:      $A.solution[a_i] \leftarrow \text{MPC}(M, a_i, A.constraints[a_i])$ 
35:      $A.cost \leftarrow \text{SIC}(A.solution, G_r)$ 
36:      $T_s.insert\_node(A)$ 
37:   else ▷ Robot-Robot Collision
38:      $C\_agents \leftarrow [a_i, a_j]$ 
39:     for  $i \leftarrow 0$  to  $\text{len}(C\_agents) - 1$  do
40:        $a_i \leftarrow C\_agents[i]$ 
41:        $A \leftarrow \text{Node}(\epsilon_r, \epsilon_o, D)$ 
42:        $A.solution \leftarrow P.solution$ 
43:        $A.obstacles \leftarrow P.obstacles$ 
44:        $A.constraints \leftarrow P.constraints$ 
45:        $a_j \leftarrow C\_agents[i - 1]$ 
46:        $x\_solution\_aj \leftarrow P.solution[a_j]$ 
47:        $A.constraints[a_i].append(0, x\_solution\_aj, C.collision\_timesteps)$ 
48:        $A.solution[a_i] \leftarrow \text{MPC}(M, a_i, A.constraints[a_i])$ 
49:        $A.cost \leftarrow \text{SIC}(A.solution, G_r)$ 
50:        $T_s.insert\_node(A)$ 
51:     end for
52:   end if
53: end while

```

5.1 Simulation Environment

The simulation was conducted using two different approaches. First, the model was implemented in a simplified 3D environment that mimics a warehouse. This environment was simulated using the StageROS package within the Robot Operating System (ROS) Noetic, as illustrated in Figure 5.1. The SLAM-Gmapping package was utilized to obtain laser data from this environment, which was then converted to obstacles using the Hough transform[19].

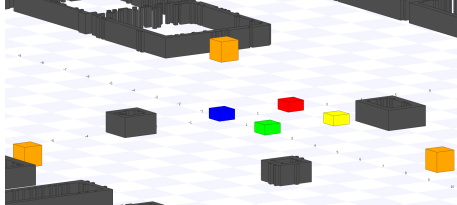


Figure 5.1: Stage 3D simulation

To further evaluate the environment, a Python-based script was developed to simulate randomly generated environments with randomly spawned humans and robots. This allowed for comprehensive testing of the algorithm under various scenarios, scales, and distributions of robots and humans.

At the core of the simulation is the process of solving Non-Linear Optimization Problems (NLP) within each prediction horizon. The CasaDi framework was used to implement Interior Point Optimization (IPOPT), which efficiently handles these newly formulated NLP problems in each sliding horizon.

The parameters for the social force simulation, which were used to model human dynamics, are listed in Table 5.1. These parameters were adapted from the original work of Helbing et al.[12], with slight modifications to suit the simulation environment.

Parameter	Value	Description
τ	0.5 s	Relaxation time
s_β	0.2 m	Step width
R_α	0.35 m	Human radius
$\frac{v_\alpha^{\max}}{v_\alpha^0}$	1.35	Maximum speed multiplier
2ϕ	200°	Field of view
ϵ_g^s	0.1 m	Goal threshold
ϵ_o^s	0.2 m	Obstacle threshold
$V_{\alpha\beta}^0$	2.1 m ² s ⁻²	Maximum interpersonal threshold
σ	0.3	Standard deviation of velocity distribution
$U_{\alpha B}^0$	10 m ² s ⁻²	Maximum border threshold
R	0.2 m	Equipotential circle radius

Table 5.1: Social force model simulation parameters

The parameters for the Human-Aware MPC were derived by adjusting values from the original work on multi-robot planning with static obstacles by Tajbaksh et al.[14] to fit a dynamic scenario. The parameters that yielded the best results in this study are provided in Table 5.2

5.2 Test Scenarios

Extensive testing has been conducted to evaluate the capabilities of the algorithm.

1. To study the Effect of environmental complexity, A constant human and robot configuration was tested against increasing environmental complexity.
2. Effect of environment scaling was explored by varying the scale of simulation as shown in Figure 5.2
3. Fleet scaling was studied by varying the number of robots across test cases.
4. Influence of crowding was examined by varying human presence across test cases.

Range of variation of all settings are shown in Table 5.3.

5.3 Evaluation Metrics

To understand robot-level performance, the Average fleet solving time, and the maximum time taken by any robot in the fleet to solve a given configuration were measured. In addition, the make span, which is the time taken by the entire

Parameter	Value	Description
Q	diag[120, 120]	Reference error coefficient
P	diag[12.5, 12.5]	Goal error coefficient
R	diag[12.0, 12.0]	Control effort penalty
N	30	Horizon length
D	1.04 m	Robot diameter
D_{HSA}	1.0 m	Human safety area diameter
ϵ_g	0.02 m	Goal tolerance
ϵ_o	0.05 m	Obstacle collision tolerance
ϵ_r	0.08 m	Inter-robot collision tolerance
T	0.2 s	MPC time step
k_h	10^8	HSA violation slack coefficient
k_r	10^6	Inter-robot collision slack coefficient
k_o	10^6	Obstacle collision slack coefficient

Table 5.2: MPC simulation parameters

Parameter	Range	Description
N_a	2-10	Number of robots
N_h	0-10	Number of humans
<i>Occupancy</i>	0-35	Percentage obstacle occupancy

Table 5.3: Evaluation parameters

fleet to complete the assigned task, was also observed. Due to the limitations of the simulated environment, human-related metrics were not recorded during this study.

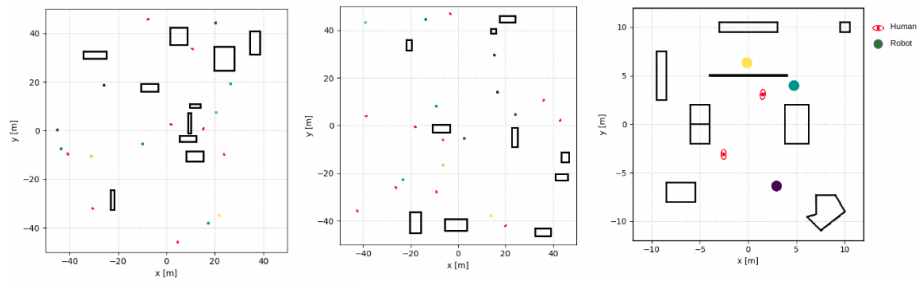


Figure 5.2: Simulations in the Python-based custom simulator at varying scales and complexity

6.1 Results

This section presents the results of the experimental evaluation of the HAM-CBS algorithm. The performance metrics include solving time per robot, fleet makespan, and average and maximum solution times under varying conditions.

Figure 6.1 represents the variation in per-robot solution time T_{avg} and maximum solution time T_{max} at various levels of environmental complexity. This simulation was set up with two robots and two humans in a small area. There is clearly an increasing solving time with increasing occupancy percentage.

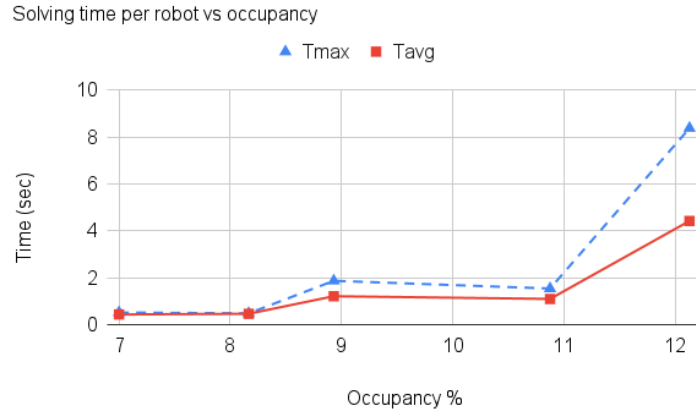


Figure 6.1: Per robot solution time vs environmental occupancy

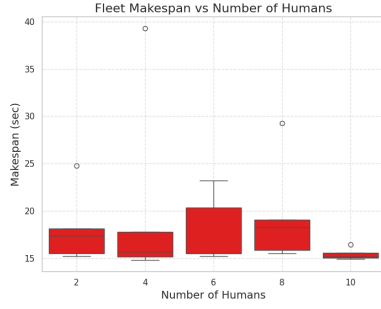


Figure 6.2: Fleet makespan vs number of humans

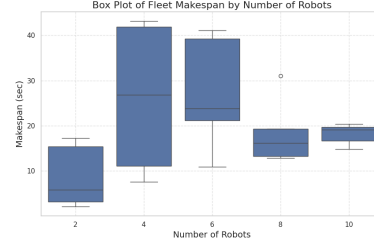
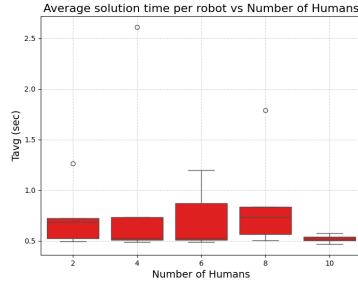
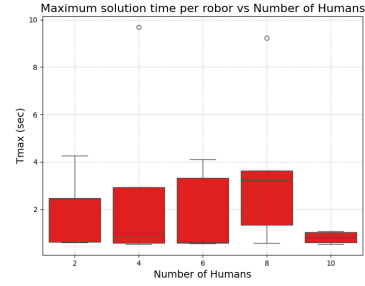


Figure 6.3: Fleet makespan vs number of robots

Figures 6.2,6.3 show the variation in makespan, the time taken for all robots to complete the tasks vs number of humans, and number of robots, respectively. Each case was simulated while keeping the other constant. The number of humans was changed while keeping 10 robots, and the number of robots varied while keeping three humans.



(a) Average solution time per robot vs Number of humans



(b) Maximum solution time per robot vs Number of humans

Figure 6.4: Comparison of average and maximum solution time per robot as the number of humans increases.

The change in solution time with respect to the number of robots is represented in Figures 6.4(a),6.4(b). The solution time complexity appears to be stochastic owing to the randomized nature of the environment. There seems to be no drastic influence of computational complexity on the increasing number of humans.

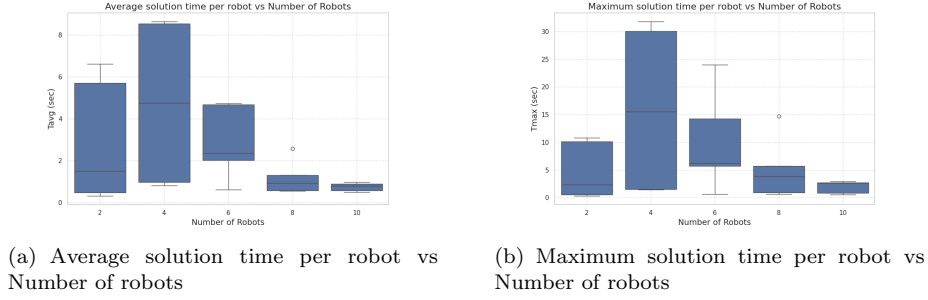


Figure 6.5: Comparison of average and maximum solution time per robot as the number of robots increases.

When the number of robots is increased, average as well as maximum solving time seem to decrease after the initial increase, as shown in Figure 6.5. This trend is explained only by the simulation itself, where a change in the nature of the global plan occurred. The only feasible goal assignments possible after adding more and more robots were for short distances.

6.2 Comparison with Other Methods

A direct comparison is not possible with the non-human aware planners. A position-swapping configuration of a varying number of robots was tested in an open environment without humans. Figure 6.6 shows that HAMCBS is able to solve the configuration faster, if not at the same time, when compared with Proritizes-MPC(Pr-MPC)[14]local planner.

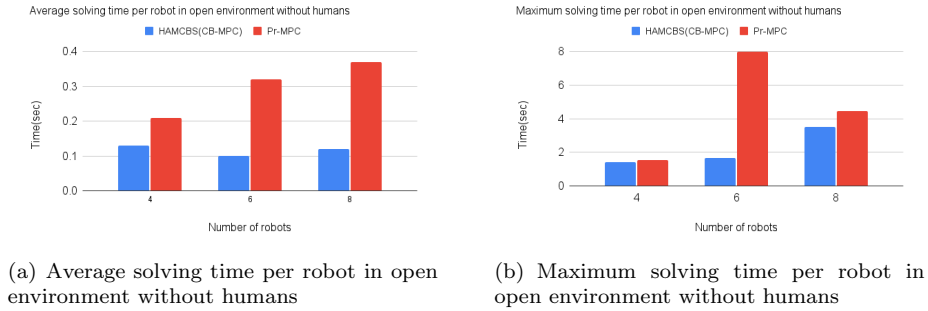


Figure 6.6: Comparison of average and maximum solving time per robot in an open environment without humans.(lower is better)

6.3 Discussion

The HAMCBS algorithm is very scalable, as increasing the number of robots or humans or the total size of the environment did not result in increased solving time,s as shown in Figures 6.2 through 6.5. However, from analysing the

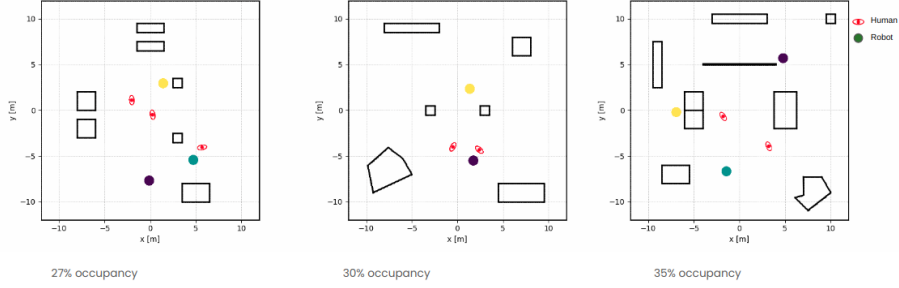


Figure 6.7: Solving environments with high occupancy percentage

algorithm performance at various occupancies, the performance drop is significant with increasing obstacle percentage. The algorithm did manage to solve high occupancy settings as illustrated in Figure 6.7, with a high solving time of 276.38 seconds. When compared to Pr-MPC, HAMCBS does have a better time complexity.

Taking a closer look at the effect of occupancy in Figure 6.1, It is well within expectations to have increasing solution times with increasing complexity since more conflicts will arise and hence more number of iterations will be needed to find a solution. It is also notable that the difference between maximum and average is increasing, which implies that while solutions to most cases are found, in these cases, there is a very small fraction of robots stuck in more complex areas.

When comparing the make-spans, we can understand that the nature of the results obtained when varying humans is probably due to the fact that humans repel each other, and hence, populating with more people in a sparse environment might make no difference to the number of humans interfering with the robot. On the other hand, the trend vs robots is more like what we would expect, even though the mean is not in a strictly increasing fashion, it does seem to follow that profile. This can be simply explained since a linear increase in make span should be expected as the number of robots increases.

In the last two cases, we can observe a low variance towards the end. This is probably due to most configurations being closer to a uniform distribution due to the high number of humans or robots. Changing the number of humans does not seem to have a definite effect on solution complexity from Figure 6.4. The variation can only be attributed to the stochastic nature of the environment.

Number of robots, on the other hand, clearly shows a large variety in problem complexity when there is more free space per robot, making various global plans possible as the number of robots increases, simpler, more uniform situations are created.

The core objective of this study was to impart human awareness to multi-robot motion planning systems. The proposed algorithm has been shown to be able to successfully solve a variety of randomized environments.

6.4 Limitations

Coming to the limitations of this work, the increasing time complexity with respect to occupancy needs to be addressed. Furthermore, although each planning horizon is optimal, total optimality of the plan is not evaluated during this study.

Lack of coverage in robot control in a physics-based simulation addressing kinematic and dynamic uncertainties is also a drawback.

Finally, estimation of factors representing human comfort or psychic state would greatly improve validation of the project's aim.

CHAPTER 7

Conclusion

7.1 Key Contributions

This work serves as a starting point for integrating social models into planning methodologies. During the scope of this work,

- A human dynamics prediction model was successfully developed.
- A Human-aware MPC local planner was formulated and implemented.
- Thorough evaluation of robot robot-level algorithm performance was conducted.

It can be said that the project achieved its original goal of developing a motion planning system integrating human dynamics prediction, dynamic obstacle avoidance, and multi-robot planning successfully.

7.2 Future Work

The main motivation for this work was a safe and effective environment for humans in human-multi-robot collaborative workspaces. This project could not evaluate human safety or satisfaction metrics, which would be the immediate future work. Replacing the computationally expensive NLP solver with a MARL neural network can bring speed and adaptability to the system. Better evaluation and improved solver configurations can potentially develop this work to a deployable level in industrial settings in future iterations.

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APPENDIX A

Instructions to run the implementaion

The implementation is uploaded in gitLab.

Location 1

<https://git.cs.bham.ac.uk/projects-2023-24/mxk555>

Location 2

HTTP :

<https://git.cs.bham.ac.uk/mxk555/human-aware-multi-robot-motion-planning.git>

SSH:[git@git.cs.bham.ac.uk:mxk555/human-aware-multi-robot-motion-planning.git](https://git.cs.bham.ac.uk/mxk555/human-aware-multi-robot-motion-planning.git)

Location 3

https://github.com/saleeqmohammed/human_aware_MRMP.git

- Run the python script `simulation_random.py` in `src` folder to simulate a randomly generated environment.
- Run `simulation.py` to run the laser scan environment.
- The core algorithm is in `./src/hacbs/planning.py`
- to run ROS version please install, ROS noetic, `ros-noetic-stageros`, `ros-noetic-slam-gmapping`, `python-opencv`. clone the repository to `src` folder in catkin workspace and run `catkin_make`

Number of Robots	Makespan (sec)	T_{avg} (sec)	T_{max} (sec)
2	5.8512	1.4967	2.3727
2	17.2925	6.6188	10.1336
2	15.4026	5.6965	10.7940
2	2.1305	0.3078	0.3227
2	3.2275	0.4586	0.5119
4	43.1438	8.5039	29.5145
4	41.3926	8.6404	31.8030
4	12.2209	1.0044	1.5667
4	7.6408	0.8009	1.4284
6	41.0556	4.6776	14.2597
6	39.2449	4.7410	24.0145
6	21.1237	2.0064	6.2574
6	10.8781	0.5961	0.6395
6	23.7607	2.3448	5.7497
8	13.2640	0.5452	0.6061
8	31.0765	2.5730	14.7302
8	19.2693	1.2916	5.7427
8	16.1491	0.9312	3.8440
8	12.8970	0.5562	0.8891
10	16.6536	0.5742	0.8807
10	19.6596	0.8900	2.6620
10	14.8612	0.4901	0.5607
10	20.4377	0.9697	2.9540
10	19.1233	0.7877	2.7469

Table A.1: Performance metrics for different numbers of robots.

Number of Humans	Makespan (sec)	T_{avg} (sec)	T_{max} (sec)
2	24.7418	1.2665	4.2516
2	15.2346	0.4921	0.5857
2	17.3700	0.6862	2.4646
2	18.1231	0.7235	2.4233
2	15.5003	0.5249	0.6082
4	39.2918	2.6109	9.6894
4	17.7867	0.7323	2.9303
4	15.1456	0.5102	0.9774
4	15.6866	0.5219	0.5658
4	14.7982	0.4875	0.5318
6	20.3357	0.8732	3.3177
6	15.2233	0.4897	0.5455
6	15.5001	0.5234	0.6119
6	15.5254	0.5083	0.5741
6	23.2056	1.1993	4.1073
8	19.0524	0.8338	3.6366
8	18.2657	0.7342	3.2037
8	15.8471	0.5628	1.3305
8	29.2439	1.7870	9.2214
8	15.5024	0.5017	0.5751
10	16.4116	0.5775	1.0650
10	15.0868	0.5246	1.0190
10	15.2361	0.5130	0.6174
10	14.9129	0.4681	0.5293

Table A.2: Performance metrics for different numbers of humans.